

Vague by Design: Central Bank Communication with an Inattentive Public*

Federico Innocenti

Tsung-Hsien Li

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Abstract

What is the optimal communication policy for a central bank to inform an inattentive public? We develop a model of central bank communication within a Bayesian persuasion framework, incorporating information processing costs and heterogeneous attention budgets across individuals. The central bank seeks to stabilize the economy by managing inflation expectations in response to fundamental shocks. When the shock is larger than the policy response, it is optimal for the central bank to remain deliberately vague, leveraging inflation surprises to mitigate fluctuations in unemployment. When the shock is smaller than the policy response, the central bank communicates precisely, balancing accuracy against audience reach. We further examine how belief heterogeneity, the share of inattentive individuals, adaptive expectations, and skewed attention budget distributions influence the central bank's optimal communication. Our findings underscore its strategic role as an effective instrument for macroeconomic stabilization.

Keywords: Central Bank Communication, Bayesian Persuasion, Rational Inattention, Inflation Expectations, Monetary Policy, Macroeconomic Stabilization

JEL Classifications: D83, E52, E58, E61

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1 Introduction

Central banks have increasingly relied on various forms of communication to shape inflation expectations (Blinder, Ehrmann, de Haan, and Jansen, 2024; Casiraghi and Perez, 2022). Forward guidance is one such tool, through which central banks signal the future path of monetary policy and seek to anchor longer-run expectations (McKay, Nakamura, and Steinsson, 2016). In the short run, a central bank can instead complement its existing monetary policy rule by communicating how it intends to achieve its policy objectives and thereby manage near-term inflation expectations. The effectiveness of such communication depends on a trade-off between precision and audience reach: detailed policy reports provide richer information but are harder to process, whereas public speeches are easier to understand but less precise (McMahon and Naylor, 2023).

Several informational frictions shape this trade-off and limit the extent to which central bank communication is processed by the broader public. Information rigidities, whether arising from rational inattention or sticky information, constrain individuals' ability to acquire and update information (Sims, 2003; Mankiw and Reis, 2002; Coibion and Gorodnichenko, 2015), whereas limited financial literacy restricts their ability to process complex policy messages (Lusardi and Mitchell, 2023). Survey evidence documents substantial disagreement in inflation expectations across consumers and experts (Mankiw, Reis, and Wolfers, 2003), and Cornand and Hubert (2022) show that information frictions vary systematically across agent types.¹ Motivated by these findings, this paper develops a theoretically grounded account of how central banks should calibrate communication precision when audiences differ in attentiveness, information-processing capacity, and beliefs.

To this end, we study a partial equilibrium model of central bank communication that combines Bayesian persuasion (Kamenica and Gentzkow, 2011) with limited attention. The economy can be in one of two states, *weak* or *strong*, and the central bank and individuals may hold different prior beliefs about these states. In each state, the economy is hit by a shock, corresponding to either rising or falling unemployment, that moves it away from its steady state. The central bank seeks to stabilize the economy by minimizing inflation and unemployment gaps.

The central bank operates under a precommitted monetary policy rule. This rule

¹Event-study evidence shows that FOMC announcements have little direct effect on household inflation expectations (Lamla and Vinogradov, 2019), consistent with our conjecture that households may not fully process complex information. The Federal Reserve's communication during the 2021–2023 inflation episode illustrates the difficulty of conveying complex and evolving inflation dynamics to the public. See Bianchi, Faccini, and Melosi (2023) for a discussion of the drivers and persistence of inflation during this period.

determines actual inflation and therefore pins down the inflation gap, leaving it outside the influence of communication. Following the classic Phillips curve, the unemployment gap depends on the inflation surprise, defined as the difference between actual and expected inflation, through the sensitivity of unemployment to unexpected inflation. Communication can therefore serve as a complementary policy tool: by strategically choosing the precision of its message and committing to a state-contingent communication strategy, the central bank shapes individuals' posterior beliefs and inflation expectations. Through the resulting inflation surprise, communication helps stabilize the unemployment gap in response to unemployment shocks.

We distinguish between attentive and inattentive individuals, corresponding in a stylized way to institutional investors and households, respectively. Attentive individuals process information costlessly, whereas inattentive individuals face information-processing costs. Following [Sims \(2003\)](#), we model these costs using entropy: messages that induce larger belief updates are more costly to process. Each inattentive individual has an idiosyncratic attention budget. If the complexity of the central bank's communication exceeds this budget, the individual ignores the message and forms inflation expectations based solely on prior beliefs. The central bank therefore faces a trade-off between precision and audience reach: more precise communication better aligns individuals' posterior beliefs with the state but may be too complex to reach a broad audience, whereas less precise communication is more accessible but less informative.

As our benchmark, we assume homogeneous neutral priors, symmetric shocks, rational expectations, and a uniform distribution of attention budgets. Under these assumptions, we theoretically characterize the optimal central bank communication, which exhibits two distinct regimes along the extensive margin. When the shock magnitude exceeds the effective power of inflation surprises, defined as the product of the Phillips curve sensitivity and the inflation surprise, the central bank optimally communicates uninformatively to preserve full inflation surprise. Doing so helps offset the adverse effects of the shocks and stabilizes the economy. Otherwise, the central bank communicates informatively to align expectations with actual inflation, mitigating excessive inflation surprises. At the intensive margin, however, precision is constrained by the share of inattentive individuals. The larger this share, the more the central bank must trade precision for audience reach, making full revelation suboptimal. In other words, the central bank communicates more precisely when its audience consists primarily of institutional investors rather than households.

We then examine the role of belief heterogeneity in central bank communication. When individuals have non-neutral priors, one state becomes subjectively more plausible. As a result, inflation surprises diminish in the more plausible state but grow increasingly large in the less plausible one as individuals assign greater probability to

the former. At the extensive margin, sufficiently non-neutral individual priors make informative communication optimal even for shocks that would call for vagueness under neutral priors, because the inflation surprise in the state individuals deem less likely becomes excessive relative to the shock.

At the intensive margin, when inflation surprises are excessive, the central bank focuses on reducing the inflation surprise in the state individuals deem more plausible, since messages that reinforce existing beliefs are simpler, and thus reach a broader audience. As messages jointly determine posterior beliefs, it does so by communicating precisely about the less plausible state, indirectly driving beliefs about the more plausible state toward one. When priors become sufficiently extreme, however, the inflation surprise in the more plausible state shrinks, and correcting it further would be counterproductive. The central bank then turns to the excessive inflation surprise in the less plausible state, communicating precisely about the more plausible state to drive beliefs about the less plausible one toward zero.

When the central bank has non-neutral priors, we find that, at the extensive margin, it switches from uninformative to informative communication only when its beliefs diverge sufficiently from those of individuals. At the intensive margin, the central bank's priors determine the relative urgency it assigns to managing inflation surprises across states. Accordingly, it is optimal to be more precise about the state the central bank considers more likely. To preserve audience reach among inattentive individuals, the central bank accordingly reduces precision for the state it considers less plausible.

The rational expectations hypothesis has been increasingly questioned. When individuals form adaptive inflation expectations that are partially anchored to their priors, this inertia weakens the effectiveness of communication. The central bank optimally responds by communicating more precisely to mitigate information loss during individual belief updating. The greater the degree of adaptiveness, the greater the scope for the central bank to deploy informative communication.

As the attention budget distribution becomes more right-skewed, that is, as the share of low-budget inattentive individuals increases, the central bank optimally reduces communication precision to maintain broader engagement. However, this logic reverses when a sufficiently large share of fully attentive individuals coexists with a predominantly low-budget inattentive population. In that case, the central bank finds it optimal to abandon the inattentive group (e.g., households) altogether, sending fully informative messages targeted exclusively at attentive individuals (e.g., institutional investors).

Related Literature. This paper contributes to the literature on central bank communication, which has traditionally relied on cheap talk models, in which the central bank lacks commitment to its communication strategy (Crawford and Sobel, 1982). Notable contributions include Stein (1989), Moscarini (2007), and Bassetto (2019). By contrast, we assume full commitment, placing our paper within the Bayesian persuasion literature (Kamenica and Gentzkow, 2011).² In such a setting, Herbert (2022) shows that countercyclical communication is optimal under coordination externalities and belief heterogeneity. By contrast, we do not consider coordination externalities but rather focus on the role of central bank communication to shape inflation expectations. Ko (2023) considers a central bank that receives a private, imperfect signal about the state of the economy and chooses how much information to disclose to fully attentive agents. In contrast, in our setting, the central bank can provide perfectly informative messages about the state of the economy, but doing so may exceed the cognitive capacity of inattentive individuals (Sims, 2003).³

Financial markets have long been studied as coordination games. Morris and Shin (2002) show that more precise public information can reduce welfare when coordination motives lead agents to overweight it relative to private information. Angeletos and Pavan (2007) clarify that this result hinges on the equilibrium degree of coordination being inefficiently high, and more generally characterize when public information is welfare-improving. Building on Morris and Shin (2002), Cornand and Heinemann (2008) show that precision should be maximal, while the share of agents receiving the information may optimally be restricted. Relatedly, Baeriswyl and Cornand (2014) show that a central bank can curb overreaction to its disclosures by limiting either their reach (partial publicity) or their precision (partial transparency). The optimal response to a cost-push shock depends on the disclosure regime (Baeriswyl and Cornand, 2010), whereas greater opacity can be optimal when the central bank’s information is endogenous to the market it shapes (Baeriswyl, Cornand, and Ziliotto, 2020). Unlike this strand of literature, we abstract from coordination externalities and instead study communication as a tool for managing inflation surprises. Moreover, the share of agents reached is not chosen directly by the central bank but arises endogenously from rational inattention: more precise messages are more complex, and inattentive individuals process them only when they fall within their attention budgets.

²Bayesian persuasion has also been used to study the Delphic forward guidance puzzle (Fujiwara and Waki, 2022). A related literature examines central bank communication under learning-based frameworks (Eusepi and Preston, 2010; Cornand and Dos Santos Ferreira, 2025). Differently from these papers, we endogenize the precision of communication.

³A theoretical literature studies persuasion of receivers with costly attention (Bloedel and Segal, 2020; Lipnowski, Mathevet, and Wei, 2020, 2022; Wei, 2021; Matyskova and Montes, 2023). Relative to it, we embed the attention constraint in a macroeconomic stabilization problem in which the central bank communicates with attentive and inattentive receivers.

Gaballo (2016) studies the limits of forward guidance when agents subjectively interpret the central bank’s communication. Agents rely heavily on the current market price to forecast the future, amplifying shocks and eroding the informational value of prices. Increasing the precision of the announcement can paradoxically worsen welfare because expectations become more reactive to news about future fundamentals, raising price volatility and agents’ *ex ante* uncertainty.⁴ Our paper differs in two ways. First, we assume the central bank optimally designs communication, whereas in Gaballo (2016) precision is exogenous. Second, in our framework vagueness is a deliberate instrument: by communicating uninformatively, the central bank strategically preserves inflation surprises that help offset shocks, turning opacity into a tool for stabilization.

We organize the rest of the paper as follows. Section 2 outlines the model. Section 3 characterizes our benchmark results. Sections 4 and 5 examine the role of belief heterogeneity, attention budgets, and irrational expectations. Section 6 concludes. We provide all proofs in Appendix A, and we present an extension on shock magnitude and asymmetry in Appendix B.

2 Model

The economy consists of a central bank and a unit-measure continuum of individuals. We define the state space as $\Omega := \{\omega_1, \omega_2\}$. The state of the economy $\omega \in \Omega$ represents an employment shock and can take on two values: when the economy is *weak*, $\omega_1 > 0$, unemployment rises; when the economy is *strong*, $\omega_2 < 0$, it falls. Individuals share a common prior belief $\mu_0 \in [0, 1]$ that state ω_1 will occur, whereas the central bank holds its own prior belief $\mu_0^c \in [0, 1]$. Accordingly, the beliefs assigned to state ω_2 are $1 - \mu_0$ for individuals and $1 - \mu_0^c$ for the central bank.

The central bank faces the following classic expectations-augmented Phillips curve:⁵

$$(u - u^N) = \omega - \gamma(\pi - \pi^e), \quad (1)$$

where u denotes the unemployment rate, u^N the natural unemployment rate, $\gamma > 0$ the sensitivity of the unemployment gap to inflation surprises, π actual inflation, and π^e expected inflation. Equation (1) describes how the unemployment gap, $u - u^N$, responds to inflation surprise, $\pi - \pi^e$. In particular, $\gamma(\pi - \pi^e)$ reflects the extent to which the inflation surprise can offset the employment shock ω .

⁴Agents’ beliefs inherently contain both a fundamental and a non-fundamental component, that cannot be separately identified, as formalized by Chahrour and Jurado (2018), who show that news and noise representations of agents’ beliefs are observationally equivalent.

⁵This form is standard. See also the accelerationist Phillips curve in Hazell, Herreño, Nakamura, and Steinsson (2022).

The central bank manages actual inflation π according to the following state-contingent monetary policy rule, referred to as the Qvigstad rule (Qvigstad, 2005):⁶

$$\pi(\omega, \pi^T, \nu) := \begin{cases} \pi^T + \nu & \text{if } \omega = \omega_1, \\ \pi^T - \nu & \text{if } \omega = \omega_2, \end{cases} \quad (2)$$

where π^T denotes the inflation target and ν represents state-dependent monetary policy flexibility.⁷ According to Equation (2), the central bank raises inflation above its target in the weak state to stimulate economic recovery and lowers it below its target in the strong state to preempt inflationary pressure.

The goal of the central bank is to stabilize the economy by minimizing quadratic losses across states arising from unemployment and inflation gaps. The central bank's objective function is $L(\omega, \pi^T, \nu)$, defined as follows:⁸

$$L(\omega, \pi^T, \nu) := (u - u^N)^2 + \alpha (\pi - \pi^T)^2 \quad (3)$$

$$= \left[\omega - \gamma (\pi(\omega, \pi^T, \nu) - \pi^e) \right]^2 + \alpha (\pi(\omega, \pi^T, \nu) - \pi^T)^2, \quad (4)$$

where $\alpha \geq 0$ denotes the relative weight the central bank places on the inflation gap relative to the unemployment gap.⁹ Equation (4) follows from substituting equations (1) and (2) into equation (3).

The timing is as follows. Before the state of the economy is revealed, the central bank commits to an information structure σ . When a shock is detected, the central bank privately infers it and communicates according to its commitment, after which individuals form posterior beliefs $\mu \in [0, 1]$ about the state and inflation expectations $\pi^e(\mu)$. Once the shock's macroeconomic effects materialize, the central bank sets its monetary policy instrument conditional on the realized state.

In the long run, the central bank could adjust its inflation target π^T or alter the flexibility ν of its monetary-policy rule. However, in the short run, doing so would undermine its credibility. The central bank can instead rely on communication to deliberately shape inflation expectations π^e as a timely stabilization tool. This paper focuses on the short run. In other words, we treat π^T and ν as exogenously given and,

⁶See also Walsh (2014), Svensson (2014) and Kahn (2015). Ko (2023) adopts this rule to study optimal central bank forward guidance.

⁷For instance, ν can be interpreted as the degree of liquidity expansion or money velocity adjustment used to accommodate inflation.

⁸This parsimonious specification admits welfare-theoretic microfoundations (Woodford, 2003).

⁹When $\alpha = 1$, the central bank places equal weight on stabilizing the inflation and unemployment gaps. $\alpha > 1$ indicates a relatively greater weight on inflation stabilization, whereas $0 \leq \alpha < 1$ indicates a relatively greater weight on unemployment stabilization.

for notational simplicity, omit them as function arguments. Since the inflation gap is pinned down by the precommitted monetary policy rule and is therefore irrelevant for information design, we omit it from the objective loss function;¹⁰ the unemployment gap is thus the sole margin available for stabilization. Thus, the central bank's ultimate goal is for the employment shock and the inflation surprise effect on unemployment to have opposite signs, possibly compensating for each other and minimizing the unemployment gap. The central bank minimizes the unemployment gap under its own prior, taking as given the inflation expectations formed by individuals. We denote the inflation surprise under state ω and belief μ by $\Pi(\omega, \mu) := \pi(\omega) - \pi^e(\mu)$. Thus, the central bank's objective becomes:

$$\mathbb{E}_{\mu_0^e} L(\mu, \sigma) = \mu_0^e [\omega_1 - \gamma \Pi(\omega_1, \mu)]^2 + (1 - \mu_0^e) [\omega_2 - \gamma \Pi(\omega_2, \mu)]^2. \quad (5)$$

The information structure σ consists of a set of messages $S := \{s_1, s_2\}$ and a family of conditional distributions $\{\sigma(\cdot | \omega)\}_{\omega \in \Omega}$ over S , that is, $\sigma : \Omega \rightarrow \Delta(S)$. Each message $s_j \in S$ sent by the central bank induces the following posterior belief μ_j about state ω_1 among individuals:

$$\mu_j = \frac{\sigma(s_j | \omega_1) \mu_0}{\sigma(s_j | \omega_1) \mu_0 + \sigma(s_j | \omega_2) (1 - \mu_0)}. \quad (6)$$

As a result, σ induces a distribution τ over posteriors μ_j , that is, $\tau \in \Delta(\Delta(\Omega))$. The probability of each message s_j (or, equivalently, each posterior μ_j) as perceived by individuals is:

$$\tau_j = \sigma(s_j | \omega_1) \mu_0 + \sigma(s_j | \omega_2) (1 - \mu_0). \quad (7)$$

The martingale property—or Bayes plausibility condition—must hold: $\mathbb{E}_\tau[\mu_j] = \mu_0$. Without loss of generality, we assume that the optimal σ satisfies $\mu_1 \geq \mu_0$ and $\mu_2 \leq \mu_0$.

Messages can be interpreted as the arguments the central bank conveys to support its state-contingent monetary policy described in Equation (2), with s_1 associated with ω_1 and s_2 with ω_2 . The conditional probability $\sigma(s_j | \omega_j)$ captures the precision of communication—that is, how strongly a message is conditioned on the underlying state. Messages are complementary, meaning that they jointly determine posterior beliefs. In particular, following Equation (6), the central bank can issue a more informative message for the weak state, that is, $\sigma(s_1 | \omega_1) \rightarrow 1$, to indirectly shape posterior beliefs about the strong state, that is, $\mu_2 \rightarrow 0$, via the law of total probability.¹¹ The

¹⁰Thus, α does not affect optimal communication here, but becomes relevant when monetary and communication policies are jointly designed. We discuss this issue in the conclusion.

¹¹Intuitively, when $\sigma(s_1 | \omega_1) \rightarrow 1$, individuals expect to receive s_2 only if ω_2 occurs. Therefore, upon

extent to which the central bank seeks to influence individual inflation expectations depends on the precision of its messages. The central bank can communicate a fully detailed contingency plan, meaning its communication is fully informative. This requires that each message is possible exclusively under its corresponding state. Specifically, $\sigma(s_1 \mid \omega_1) = 1$ and $\sigma(s_2 \mid \omega_2) = 1$ implies $\mu_1 = 1$ and $\mu_2 = 0$. Alternatively, the central bank may choose to convey its intent vaguely, e.g., omitting implementation details. Specifically, $\sigma(s_1 \mid \omega_1) = \sigma(s_2 \mid \omega_2) = \frac{1}{2}$ corresponds to uninformative communication, as it implies no update from the prior, i.e., $\mu_0 = \mu_1 = \mu_2$.

Understanding complex policy announcements is costly. Individuals can be either attentive (e.g., institutional investors) or inattentive (e.g., households), with a share $\delta \in [0, 1]$ of the population being inattentive. Attentive individuals can process any information at no cost. Each inattentive individual i is endowed with an attention budget κ_i , drawn from an atomless distribution $F(\kappa)$ with support $[0, 1]$, and processes information σ if and only if the processing cost $c(\sigma)$ is lower than the attention budget κ_i . As a result, the most informative messages are not necessarily optimal *a priori*.

The cost of processing information is defined as:

$$c(\sigma) = \chi \left[H(\mu_0) - \sum_{j \in \{1,2\}} \tau_j \cdot H(\mu_j) \right], \quad (8)$$

where $H(\cdot)$ denotes the Shannon entropy:

$$H(\mu) = -[\mu \ln(\mu) + (1 - \mu) \ln(1 - \mu)]. \quad (9)$$

When σ is uninformative—i.e., $\mu_1 = \mu_2 = \mu_0$ —the cost is zero: $c(\sigma) = 0$. When σ is fully informative—i.e., $(\mu_1, \mu_2) = (1, 0)$ —we have $H(\mu_1) = H(\mu_2) = 0$, and hence $c(\sigma) = \chi H(\mu_0)$. The scaling parameter χ is normalized as $\chi = [H(\mu_0)]^{-1}$, so that the cost of a fully informative σ equals 1. As a consequence, since $F(\chi H(\mu_0)) = 1$, the information processing cost $c(\sigma)$ spans the full support of the attention distribution.

It follows that the mass of individuals who pay attention to the central bank is $1 - \delta F(c(\sigma))$. The posterior beliefs of those inattentive individuals who do not pay

receiving s_2 , they infer it must be conditional on ω_2 . Hence, a higher $\sigma(s_1 \mid \omega_1)$ implies a lower μ_2 .

attention remain at the common prior μ_0 . Therefore, the central bank's problem is:

$$\begin{aligned} \min_{\sigma} \quad & \delta F(c(\sigma)) \left\{ \mu_0^c [\omega_1 - \gamma \Pi(\omega_1, \mu_0)]^2 + (1 - \mu_0^c) [\omega_2 - \gamma \Pi(\omega_2, \mu_0)]^2 \right\} \\ & + (1 - \delta F(c(\sigma))) \left\{ \mu_0^c \sum_{j=1}^2 \sigma(s_j | \omega_1) [\omega_1 - \gamma \Pi(\omega_1, \mu_j)]^2 \right. \\ & \left. + (1 - \mu_0^c) \sum_{j=1}^2 \sigma(s_j | \omega_2) [\omega_2 - \gamma \Pi(\omega_2, \mu_j)]^2 \right\}. \end{aligned} \quad (10)$$

In essence, the central bank determines the precision of its messages— $\sigma(s_1 | \omega_1)$ and $\sigma(s_2 | \omega_2)$ —to minimize expected losses across states. Faced with a trade-off between message precision and audience reach (i.e., the need to reach out to inattentive individuals), the central bank strategically influences the inflation expectations of individuals to manage inflation surprises and stabilize the economy.

Discussion. Our model abstracts from three dimensions. First, we focus on direct communication between the central bank and individuals, such as households and institutional investors. We therefore do not consider the strategic role that media may play as intermediaries by filtering or distorting the information transmitted by the central bank. Second, we assume that individuals know the central bank's inflation target, which may not always be the case. If individuals were uncertain about the target, inflation surprises would also depend on the misalignment between the actual inflation target and individuals' perceived target.¹² In this case, communicating the inflation target would constitute an additional instrument that the central bank could strategically use to manage inflation surprises. Third, we abstract from policy uncertainty: following Brainard's attenuation logic (Brainard, 1967), the central bank's uncertainty about the effects of its monetary policy could make the central bank more cautious and, thus, alter the optimal precision of its communication. In this paper, we focus on a single type of uncertainty (the one related to shocks) to clearly identify the central bank's optimal communication at the extensive and intensive margins.

¹²Experimental evidence suggests that communicating inflation targets can affect expectation formation (Cornand and M'baye, 2018). For survey-experimental evidence on households' responses to information about current inflation and the Federal Reserve's inflation target, see Coibion, Gorodnichenko, and Weber (2022).

3 How Should a Central Bank Communicate?

To better understand the trade-off in central bank communication between information precision and audience reach, we consider an analytically tractable benchmark under the following assumptions:

Assumption 1. *Prior beliefs are homogeneous and neutral: $\mu_0 = \mu_0^c = \frac{1}{2}$.*

Assumption 2. *Employment shocks are symmetric: $\omega_1 = -\omega_2 = \omega > 0$.*

Assumption 3. *Individuals form rational inflation expectations:*

$$\pi^e(\mu) = \mu\pi(\omega_1) + (1 - \mu)\pi(\omega_2) = \pi^T - (1 - 2\mu)v. \quad (11)$$

Assumption 4. *The attention budget distribution is uniform: $F(c(\sigma)) = c(\sigma)$.*

The central bank aims to steer the *targeted* inflation surprises $\Pi(\omega_1, \mu_1)$ and $\Pi(\omega_2, \mu_2)$, where each posterior μ_j is induced by the message s_j designed to support state ω_j .¹³ Lemma 1 characterizes the feasible range of these targeted inflation surprises that the central bank can generate through communication.

Lemma 1. *Following Equation (2) and Assumption 3, the inflation surprises are characterized by:*

$$\Pi(\omega_1, \mu) = 2(1 - \mu)v, \quad (12)$$

$$\Pi(\omega_2, \mu) = -2\mu v. \quad (13)$$

Since $\mu_1 \in \left[\frac{1}{2}, 1\right]$ and $\mu_2 \in \left[0, \frac{1}{2}\right]$, the feasible ranges of the targeted inflation surprises are:

$$0 = \Pi(\omega_1, 1) \leq \Pi(\omega_1, \mu_1) \leq \Pi\left(\omega_1, \frac{1}{2}\right) = v, \quad (14)$$

$$0 = \Pi(\omega_2, 0) \geq \Pi(\omega_2, \mu_2) \geq \Pi\left(\omega_2, \frac{1}{2}\right) = -v. \quad (15)$$

The central bank strategically designs communication to manage negative and positive inflation surprises to counteract the state-dependent shocks. Notably, the largest feasible targeted inflation surprises arise under uninformative communication, where $\mu_1 = \mu_2 = \mu_0 = \frac{1}{2}$.

Under Assumptions 1–4, we can analytically characterize the optimal information structure that solves the minimization problem in Equation (10). In particular, the

¹³Mathematically, Equation (10) indicates that the central bank implicitly places greater weight on targeted inflation surprises as the corresponding message precision increases.

optimal message precision is symmetric across states and satisfies:

$$\sigma(s_1 | \omega_1) = \sigma(s_2 | \omega_2) \equiv \sigma^*. \quad (16)$$

Proposition 1 formalizes this result, and its proof is in the Appendix A.1.

Proposition 1. *Under Assumptions 1–4, the central bank’s optimal information design distinguishes two cases:*

1. If $2\omega \geq \gamma\nu$, the central bank communicates uninformatively: $\sigma^* = \frac{1}{2}$.
2. If $2\omega < \gamma\nu$, the central bank communicates informatively: $\sigma^* \in \left(\frac{1}{2}, 1\right]$, where σ^* solves:

$$1 + \left(\frac{d}{d\sigma} \ln(1 - \delta c(\sigma)) \right) \Big|_{\sigma=\sigma^*} \left(\sigma^* - \frac{1}{2} \right) = 0, \quad (17)$$

or $\sigma^* = 1$, if the share of inattentive individuals is sufficiently small.

Proposition 1 provides important insights into central bank communication. Recall that γ denotes the sensitivity of the unemployment gap to inflation surprises in Equation (1), whereas the monetary flexibility parameter ν captures how large inflation surprises can be, as shown in Lemma 1. Central bank communication is thus governed by the joint impact of γ and ν , relative to the magnitude of the employment shock ω .

First, the extensive margin—namely, whether the central bank chooses to communicate informatively or not—depends on whether there is scope for communication to stabilize the economy. When the shock is overwhelming, i.e., $2\omega \geq \gamma\nu$, the central bank opts for uninformative communication to exploit inflation surprises to counteract the shock. In this case, the central bank’s expected loss is $(\omega - \gamma\nu)^2$. In contrast, when the shock is mild, i.e., $2\omega < \gamma\nu$, the central bank communicates informatively to avoid excessive inflation surprises. Under fully informative communication, the central bank’s expected loss is $\delta(\omega - \gamma\nu)^2 + (1 - \delta)\omega^2$.

Second, when informative communication is preferred, the intensive margin—that is, communication precision—is constrained by the cost of processing information faced by individuals. Recall that $1 - \delta c(\sigma)$ denotes the mass of individuals who pay attention to central bank communication, given the information structure σ . The derivative $\frac{d}{d\sigma} \ln(1 - \delta c(\sigma)) \leq 0$ captures the rate at which inattentive individuals disengage as informational complexity increases. The share of inattentive individuals influences this disengagement rate: when δ is larger, disengagement becomes more sensitive to information complexity, leading to a steeper reduction in attention. The optimal precision $\sigma^* \in \left(\frac{1}{2}, 1\right]$ thus balances precision with audience reach.

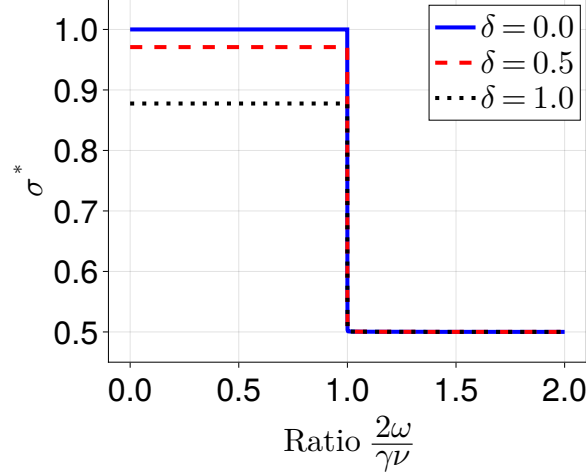


Figure 1: Optimal Central Bank Communication Strategy

We illustrate Proposition 1 in Figure 1, which displays the optimal message precision as a function of the ratio $\frac{2\omega}{\gamma\nu}$ for varying shares of inattentive individuals $\delta \in \{0, 0.5, 1\}$. The central bank switches its communication strategy at the threshold $\frac{2\omega}{\gamma\nu} = 1$. When $\frac{2\omega}{\gamma\nu} \geq 1$, the central bank opts for uninformative communication, exploiting inflation surprises to stabilize the economy. In contrast, for $\frac{2\omega}{\gamma\nu} < 1$, it communicates informatively, but precision declines with higher shares of inattentive individuals, as the central bank strategically trades off precision for broader audience reach.

Discussion. Our benchmark results highlight the critical role of central bank communication in macroeconomic stabilization. During moderate periods, the central bank can strategically deploy informative communication to attract attention and guide inflation expectations, thereby stabilizing the economy without frequently adjusting the underlying monetary policy rule. By contrast, in times of severe distress, it may be optimal for the central bank to remain deliberately vague. In such cases, uninformative communication exploits inflation surprises, helping to absorb the shock and mitigate unemployment fluctuations.

As shown in Proposition 1, the central bank's decision to communicate informatively or not at the extensive margin hinges on *the fundamental comparison* between the magnitude of the shock and the effective power of inflation surprises. When informative communication is optimal, the central bank strategically designs its communication, trading off precision against audience reach at the intensive margin. Although we derive Proposition 1 under Assumptions 1–4, its core insights—the fundamental comparison and the precision–audience reach trade-off—remain robust and provide an intuitive framework for interpreting the results when these assumptions are relaxed.

Table 1: Benchmark Parameterization

Parameter	Value	Description
δ	1	Share of inattentive individuals (households)
ω_1	1	Magnitude of unemployment shock in the weak state
ω_2	-1	Magnitude of employment shock in the strong state
μ_0	0.5	Individual prior belief
μ_0^c	0.5	Central bank prior belief
γ	2.94	Sensitivity of unemployment to inflation surprises
π^T	2	Inflation target
ν	1	Monetary policy flexibility

4 Communication under Belief Heterogeneity

In practice, the central bank and individuals may hold divergent economic outlooks, which we capture in our framework as belief heterogeneity. We relax Assumption 1 to investigate the role of such heterogeneity in shaping central bank communication. To this end, we numerically solve for the optimal communication strategy defined in Equation (10), allowing for heterogeneous priors. Section 4.1 examines how belief heterogeneity affects the central bank's optimal communication at the intensive margin. Section 4.2 explores how belief heterogeneity affects the extensive margin by considering a flatter Phillips curve.

Benchmark Parameterization. The share of inattentive individuals is set to $\delta = 1$. The shock magnitudes are symmetric, with $\omega_1 = 1$ and $\omega_2 = -1$, representing deviations of one percentage point in the unemployment rate from its natural level. Both individuals and the central bank are assumed to hold homogeneous and neutral prior beliefs, with $\mu_0 = \mu_0^c = 1/2$. The inflation sensitivity parameter is set to $\gamma = 2.94$, calibrated to match the estimate of $\psi = 0.34$ in Hazell et al. (2022).¹⁴ The central bank's inflation target is fixed at $\pi^T = 2$, reflecting the standard 2% target used in practice. The monetary policy parameter is set to $\nu = 1$, indicating that the central bank tolerates a 1 percentage point deviation from the target to stabilize the economy. Table 1 summarizes the benchmark parameter values used in the analysis.

Definition 1. We say that $\mu_0 \rightarrow 0$ or $\mu_0^c \rightarrow 0$ reflects optimism (a higher belief in the strong state ω_2), whereas $\mu_0 \rightarrow 1$ or $\mu_0^c \rightarrow 1$ reflects pessimism (a higher belief in the weak state ω_1).

¹⁴In our framework, γ corresponds to the reciprocal of the Phillips curve slope parameter ψ in Hazell et al. (2022), i.e., $\gamma = 1/\psi$.

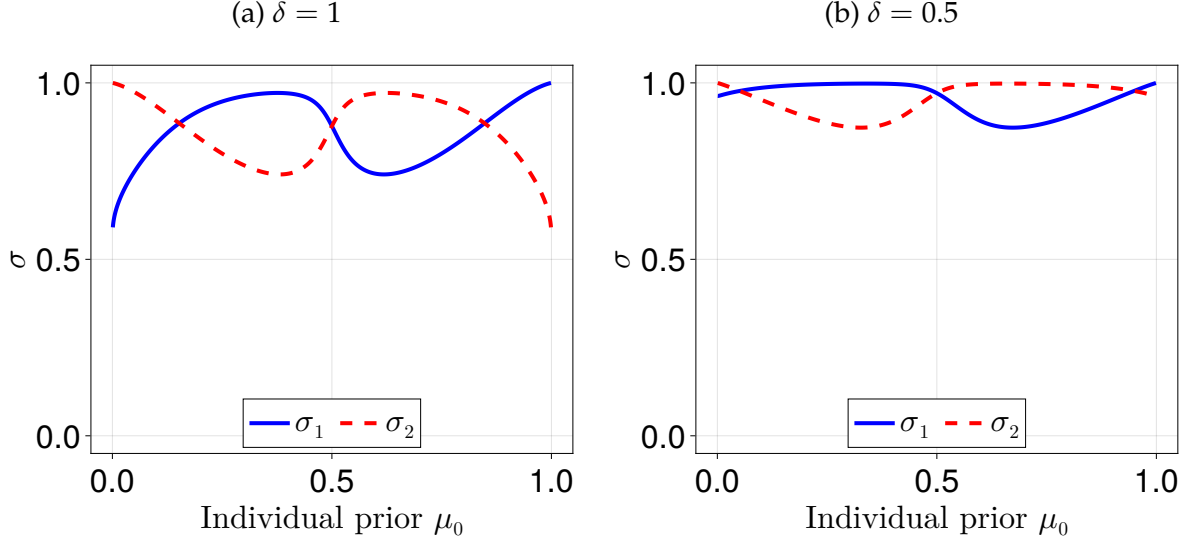


Figure 2: Optimal Message Precision Across Individual Priors

4.1 Intensive Margin

Our benchmark parameterization falls into the second case of Proposition 1, where the central bank communicates informatively and trades off precision for audience reach to optimally manage inflation surprises. In this section, we examine how belief heterogeneity affects the intensive margin. We consider two scenarios: one in which individuals' prior belief μ_0 varies from 0 to 1 while holding all other parameters fixed, and another in which the central bank's prior belief μ_0^c varies from 0 to 1 while keeping the remaining benchmark parameters constant. We also vary the share of inattentive individuals δ to explore how inattentiveness interacts with belief heterogeneity in shaping the optimal communication strategy.

Figure 2 presents the optimal message precision as a function of individual prior beliefs ranging from 0 to 1, where $\sigma_1 = \sigma(s_1 | \omega_1)$ and $\sigma_2 = \sigma(s_2 | \omega_2)$. Figure 2a shows the results when all individuals are inattentive ($\delta = 1$), while Figure 2b displays the results when half of the individuals are inattentive ($\delta = 0.5$).

As illustrated in Figure 2, individual optimism and pessimism lead to deviations from the benchmark results, exhibiting a mirror-image pattern. For brevity, we focus our discussion on the case of pessimism, with the understanding that the same logic applies symmetrically to the case of optimism.

In Figure 2a, which depicts the case where all individuals are inattentive, the central bank chooses $\sigma_1 \ll 1$ and $\sigma_2 \approx 1$ when individuals are mildly pessimistic (e.g., $\mu_0 = 0.6$). This is because pessimistic beliefs imply that individuals perceive the weak state as more likely *ex ante*. Consequently, their inflation expectations are more closely aligned with actual inflation in the weak state, resulting in smaller inflation surprises.

In contrast, the strong state is perceived as less likely, leading to amplified inflation surprises. These dynamics are evident in Equations (12) and (13): as μ_0 increases, $\Pi(\omega_1, \mu_0)$ decreases toward zero, while $\Pi(\omega_2, \mu_0)$ becomes increasingly negative. To manage these asymmetric inflation surprises while preserving the audience reach of its communication, the central bank optimally induces posteriors $\mu_1 \approx 1$ and $\mu_2 \in (0, \mu_0)$. Indeed, when individuals are mildly pessimistic, inflation surprises remain excessive under either shock. However, as shown in Equation (8), the cost of processing information increases with the distance between prior and posterior beliefs. As pessimism intensifies, it becomes cheaper to reinforce beliefs about the weak state, rather than shifting individuals toward beliefs favoring the strong state. Thus, given mildly pessimistic individual priors, it is optimal to induce $\mu_1 \approx 1$ rather than $\mu_2 \approx 0$, thus focusing on reducing excessive inflation surprises under the weak state.

However, in Figure 2a, when individuals perceive the weak state as increasingly certain (e.g., $\mu_0 \rightarrow 1$), the central bank chooses $\sigma_1 \approx 1$ and $\sigma_2 \ll 1$. As previously discussed, under such pessimistic beliefs, inflation surprises in the weak state become negligible, while those in the strong state grow excessively large. In particular, there exists a threshold value of prior belief μ_0 such that the inflation surprise is not enough to compensate for the shock under the weak state, beyond that threshold. Therefore, when individuals are too pessimistic, it is optimal for the central bank to induce $\mu_2 \approx 0$, focusing on reducing excessive inflation surprises under the strong state, while using the little inflation surprise to partially compensate for the shock under the weak state.

What happens when half of the individuals are inattentive ($\delta = 0.5$)? The incentives deriving from the optimal management of inflation surprises remain as before. However, a reduction in the share of inattentive individuals predictably attenuates the importance of information processing costs. As a consequence, the central bank can afford to raise the precision of its communication: compared to Figure 2a, the overall message precisions in Figure 2b are higher across the entire range of individual priors. Notably, when individuals hold extreme prior beliefs, the central bank optimally induces $\mu_2 \approx 0$, but can do it more often than when $\delta = 1$. In particular, the central bank adopts a fully informative communication, such that $\sigma_1 \approx 1$ and $\sigma_2 \approx 1$. The central bank leverages the heterogeneity in attentiveness across individuals when designing its communication. The presence of fully attentive individuals, who can process any messages, enables the central bank to communicate more effectively. The central bank strategically targets this group with informative messages aimed at correcting extreme priors. Meanwhile, it deliberately refrains from engaging inattentive individuals with limited attention budgets. This targeted approach enhances inflation expectation management and contributes to macroeconomic stabilization at the aggregate level.

Figure 3 presents the optimal message precision as a function of central bank prior

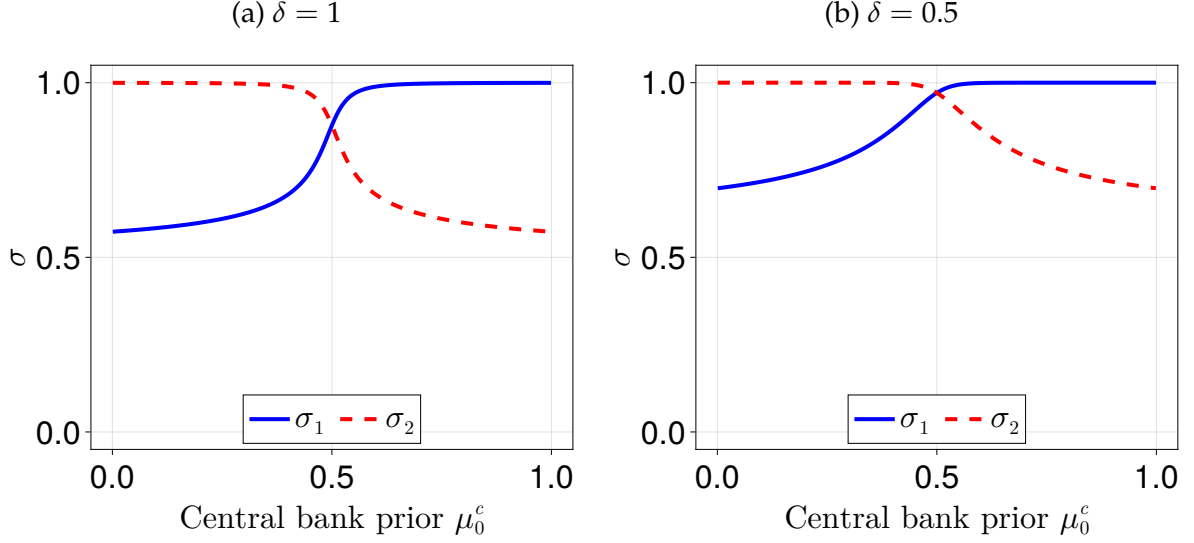


Figure 3: Optimal Message Precision Across Central Bank Priors

beliefs ranging from 0 to 1. Figure 3a shows the results when all individuals are inattentive ($\delta = 1$), while Figure 3b displays the results when half of the individuals are inattentive ($\delta = 0.5$). Again, we focus on the case of pessimism.

Recall that, in Equation (10), the central bank minimizes expected losses across states using its own prior beliefs as weights. In other words, the central bank's prior determines the relative importance it assigns to stabilizing each state.¹⁵ Based on this belief, the central bank decides whether and how to construct informative messages to guide individuals (holding neutral priors) and stabilize the economy.

In Figure 3a, when the central bank becomes more pessimistic ($\mu_0^c \rightarrow 1$), it increases σ_1 , while reducing σ_2 . In other words, the central bank optimally induces $\mu_1 \in (\mu_0, 1)$ and $\mu_2 \approx 0$. Thus, the central bank calibrates inflation surprises using message s_1 to compensate the shock under the weak state, while forgoing the management of inflation surprises under the strong state. The opposite holds when the central bank is optimistic ($\mu_0^c \rightarrow 0$). Essentially, the central bank strategically sacrifices precision in the less likely state to enhance it in the more plausible one. This trade-off reflects the central bank's effort to minimize expected losses in the state it perceives as more likely, while preserving audience reach among inattentive individuals. Figure 3b shows that reducing the share of inattentive individuals leads to an increase in precision, without altering the central bank incentives otherwise.

¹⁵This prior can be interpreted as reflecting the central bank's superior knowledge or informed judgment about the likelihood of different economic conditions.

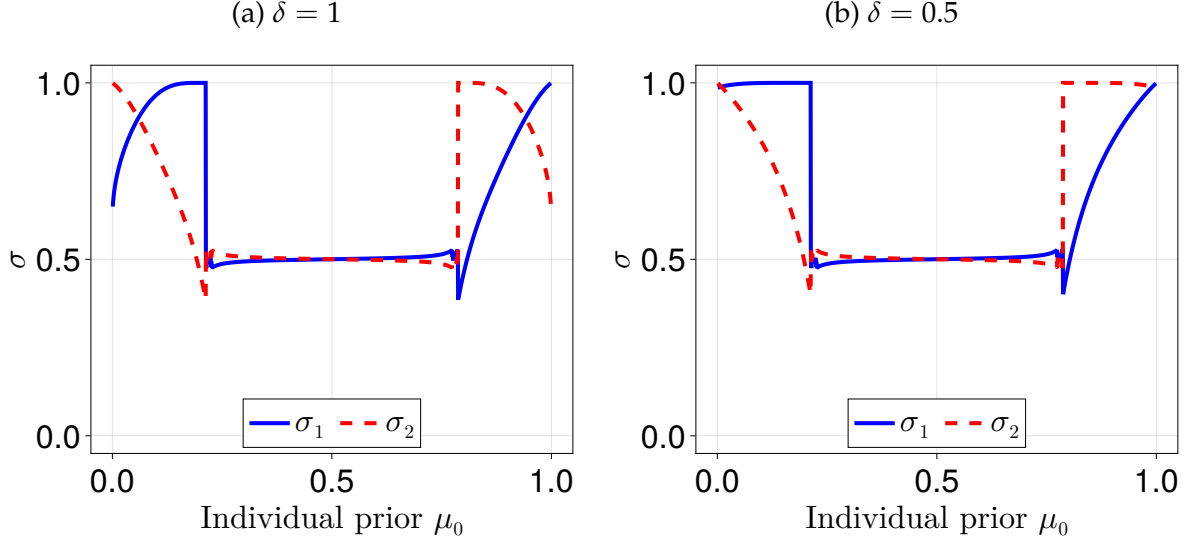


Figure 4: Optimal Message Precision Across Individual Priors when $\gamma = 1$

4.2 Extensive Margin

In this section, we investigate how belief heterogeneity interacts with the extensive margin, that is, the central bank's decision to communicate informatively or not. We do that considering a departure from our benchmark parameterization, where the Phillips curve flattens, that is, the sensitivity of the unemployment gap to inflation surprise weakens in the form of a reduced γ .¹⁶ Remarkably, the same considerations apply to settings where γ is constant but the magnitude of shocks increases. See Appendix B for a discussion.

Under the benchmark parameterization, setting $\gamma = 1$ corresponds to the first case in Proposition 1, in which the central bank optimally adopts uninformative communication to fully exploit inflation surprises for macroeconomic stabilization. Intuitively, as the power of expectation management weakens with a lower γ , and the magnitude of shocks exceeds this diminished power, the central bank resorts to vague messaging to generate the largest feasible inflation surprises. Belief heterogeneity induces asymmetric inflation surprises, thereby affecting the central bank's incentive to communicate informatively.

In Figure 4, we present the optimal communication strategy across individual prior beliefs under a flattened Phillips curve with $\gamma = 1$. Figure 4a shows the results when all individuals are inattentive ($\delta = 1$), while Figure 4b displays the results when half of the individuals are inattentive ($\delta = 0.5$).

¹⁶Empirical evidence shows that the Phillips curve has become less responsive to inflation surprises. This trend has raised concerns among policymakers regarding the diminishing effectiveness of monetary policy, particularly in its ability to shape inflation expectations for macroeconomic stabilization.

The comparison of Figure 2a and Figure 4a shows that, when $\gamma = 1$, the central bank provides information only when individuals are sufficiently optimistic or pessimistic. When individual priors are neutral and the Phillips curve is flat, the central bank remains deliberately vague, consistently with Proposition 1. Even when priors deviate slightly from one-half, the central bank continues to communicate uninformatively because the magnitude of shocks continues to exceed inflation surprises. However, once individuals become sufficiently optimistic or pessimistic, inflation surprises in the less likely state become excessive. At this point, the effective power of inflation surprises outweighs the shock's magnitude, prompting the central bank to communicate informatively. Once the central bank shifts from uninformative to informative communication, at the intensive margin it responds to beliefs heterogeneity in the same way as discussed in Section 4.1. Similarly, comparing Figure 2b and Figure 4b shows qualitatively similar results when the share of inattentive individuals decreases.

Figure 4 reveals an interesting pattern: as the Phillips curve flattens, a middle region of individual prior beliefs emerges in which uninformative communication becomes optimal for the central bank. We refer to this middle range of beliefs as the *strategic vagueness region*. The flatter the Phillips curve, the broader this region becomes, expanding the set of individual priors over which the central bank prefers to remain vague in its communication. In other words, under a flat Phillips curve, the central bank communicates informatively only when individual beliefs are excessively optimistic or pessimistic.

Following this, we offer a complementary perspective on the observed pattern by examining how the optimal communication strategy varies with central bank prior beliefs under a flattened Phillips curve ($\gamma = 1$). Figure 5a presents the case in which individuals hold neutral priors ($\mu_0 = 0.5$), whereas Figure 5b displays the results for a scenario in which individuals exhibit strong optimism ($\mu_0 = 0.1$).

Figure 5a shows that, when $\mu_0 = 0.5$, varying the central bank's prior does not affect its communication strategy at the extensive margin. Since $\mu_0 = 0.5$, inflation surprises are symmetric across states and only partially offset shocks. Changing the likelihood of these shocks for the central bank does not change how optimally it responds to them. By contrast, Figure 5b shows that, when individuals are highly optimistic, the optimal communication strategy becomes responsive to variations in the central bank's prior. When $\mu_0 = 0.1$, inflation surprises become excessively large in the weak state. If the central bank is also optimistic, it places greater weight on stabilizing losses in the strong state. Accordingly, even though inflation surprises are excessive in the weak state, the central bank intentionally focuses on minimizing losses in the strong state by maintaining uninformative communication.

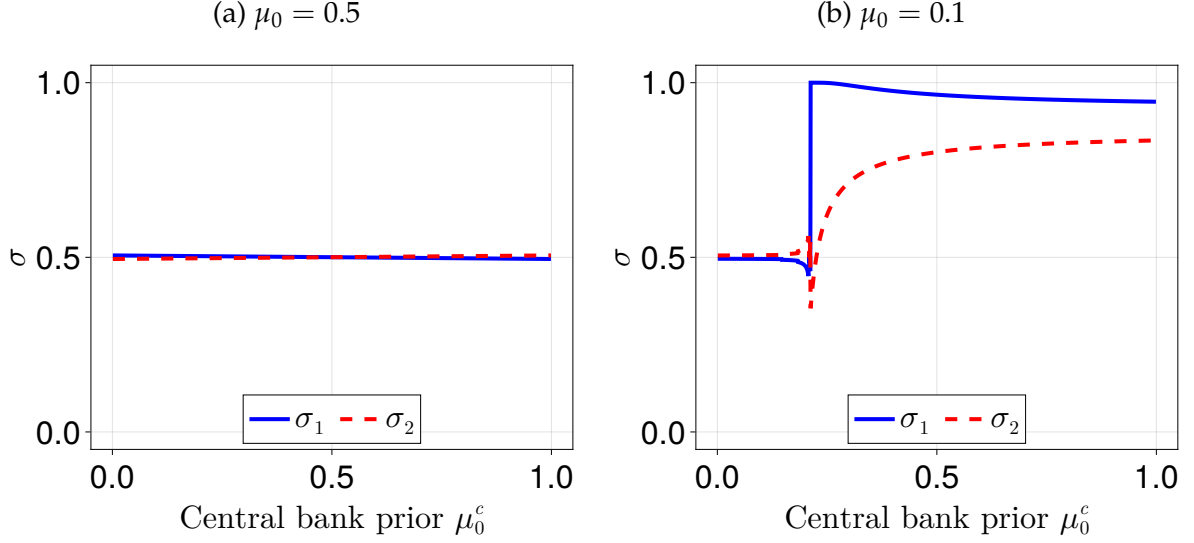


Figure 5: Optimal Message Precision Across Central Bank Priors when $\gamma = 1$

Instead, when the central bank holds a sufficiently different prior from that of individuals, the large inflation surprises in the weak state begin to factor into its decision-making. In other words, there exists a threshold in the central bank's prior belief above which it switches from uninformative to informative communication to correct the excessive inflation surprises associated with the weak state. Once this threshold is exceeded, the central bank optimally chooses $\sigma_1 \approx 1$ and $\sigma_2 < 1$ to manage excessive inflation surprises in the weak state, while strategically reducing message precision in the strong state to preserve audience reach. In particular, the optimal communication implies $\mu_2 \approx 0$ and $\mu_1 \in (\mu_0, 1)$. In other words, the central bank seeks to shift individual beliefs from highly optimistic to more pessimistic to reduce excessive inflation surprises in the weak state, that the central bank considers sufficiently likely.

5 Talking with Skewed, Behavioral Individuals

The effectiveness of central bank communication depends critically on how individuals form inflation expectations and process information. The empirical validity of the rational expectations hypothesis has been increasingly questioned, as real-world observations frequently deviate from its core assumptions. Moreover, attention budgets—or more broadly, cognitive capacities—are far from uniformly distributed across individuals. For example, a substantial portion of the population demonstrates limited financial literacy, constraining their ability to interpret complex economic information.

Thus, to further generalize our framework, we relax Assumption 3 on rational expectations and Assumption 4 on uniformly distributed attention budgets. Under

Assumption 5, we posit that individuals form *adaptive beliefs*, whereby updated beliefs under communication remain partially anchored to their priors. Under Assumption 6, we assume that attention budgets among inattentive individuals follow a Kumaraswamy distribution, which offers flexibility in capturing various forms of skewness—including the benchmark uniform distribution as a special case.

Assumption 5. *Individuals form adaptive beliefs:*

$$\hat{\mu} = \mu_0 + \theta(\mu - \mu_0) = (1 - \theta)\mu_0 + \theta\mu, \quad (18)$$

where $\theta \in [0, 1]$ governs the extent to which individuals incorporate the posterior belief μ into their updated belief $\hat{\mu}$, whereas $1 - \theta$ captures the degree of anchoring to the prior μ_0 . Notably, $\theta = 1$ corresponds to the rational expectations benchmark.

Lemma 2. *Given Assumption 5 and the linear inflation expectation function in Equation (11), individuals form adaptive inflation expectations:*

$$\pi^e(\hat{\mu}) = (1 - \theta)\pi^e(\mu_0) + \theta\pi^e(\mu). \quad (19)$$

See Appendix A.2 for the proof.

Assumption 6. *The distribution of attention budgets κ among inattentive individuals follows a Kumaraswamy distribution:*

$$f(\kappa; a, b) = a \cdot b \cdot \kappa^{a-1} \cdot (1 - \kappa^a)^{b-1}, \quad (20)$$

$$F(\kappa; a, b) = 1 - (1 - \kappa^a)^b, \quad (21)$$

where $\kappa \in [0, 1]$ and (a, b) are positive shape parameters. Notably, the case $(a, b) = (1, 1)$ corresponds to the uniform distribution.

A larger value of a shifts more probability mass toward one, producing a left-skewed distribution, whereas a larger value of b concentrates mass near zero, yielding a right-skewed distribution.

We begin by examining how Assumptions 5 and 6 modify the benchmark results presented in Section 3. To this end, we extend the theoretical framework established in Proposition 1 and derive new analytical results in Proposition 2, which characterize the optimal communication strategy when individuals form adaptive expectations and exhibit heterogeneous, skewed attention budgets. A detailed derivation of Proposition 2 is provided in Appendix A.3.

Proposition 2. *Under Assumptions 1, 2, 5, and 6, the central bank's optimal information design distinguishes two cases:*

1. If $2\omega \geq (2 - \theta)\gamma\nu$, the central bank communicates uninformatively: $\sigma^* = \frac{1}{2}$.
2. If $2\omega < (2 - \theta)\gamma\nu$, the central bank communicates informatively: $\sigma^* \in \left(\frac{1}{2}, 1\right]$, where σ^* solves:

$$1 + \left(\frac{d}{d\sigma} \ln \left(1 - \delta \left[1 - (1 - (c(\sigma))^a \right)^b \right] \right) \Big|_{\sigma=\sigma^*} \right) \left(\sigma^* - \frac{1}{2} \right) = 0. \quad (22)$$

or $\sigma^* = 1$, if the share of inattentive individuals is sufficiently small.

Compared to Proposition 1, the extensive margin now additionally depends on the degree of adaptive expectations, as captured by the term $2 - \theta$. When $\theta = 1$, we are back to rational expectations. As θ decreases, the strategic vagueness region contracts, since the condition $2\omega \geq (2 - \theta)\gamma\nu$ becomes less likely to hold, *ceteris paribus*. In other words, the presence of adaptive expectations narrows the set of circumstances under which uninformative communication is optimal. This result arises because, when $\theta < 1$, individuals form inflation expectations that remain partially anchored to their priors. This inertia in the updating process reduces the responsiveness of inflation expectations to new information. Consequently, the central bank becomes more inclined to communicate informatively to compensate for this diminished sensitivity and thus guide inflation expectations more effectively.

Recall that the derivative $\frac{d}{d\sigma} \ln(1 - \delta F(c(\sigma))) \leq 0$ captures the rate at which inattentive individuals disengage from communication as informational complexity increases. The distribution of attention budgets F also plays a critical role in shaping this disengagement rate. Following Equation (21), $F(c(\sigma)) = 1 - (1 - (c(\sigma))^a)^b$. A higher value of a shifts probability mass toward higher attention budgets, allowing inattentive individuals to tolerate greater informational complexity before disengaging. Consequently, the marginal disengagement rate flattens, reducing the audience-reach cost associated with increased precision. Conversely, a higher value of b concentrates probability mass near lower attention budgets, indicating that most inattentive individuals are more easily overwhelmed by complex messages. In this case, the disengagement rate steepens, intensifying the audience-reach cost of enhancing precision.

We next examine how the interaction between a skewed distribution of attention budgets and the share of inattentive individuals shape the optimal communication strategy under belief heterogeneity. To this end, we focus on the case of $(a, b) = (1, 5)$, which captures the empirically relevant scenario in which a substantial share of individuals possess limited attention budgets. We also vary the share of inattentive individuals: $\delta \in \{0.5, 1\}$. Figure 6 displays the results. Recall that Figure 2 reports the cases $(a, b, \delta) = (1, 1, 1)$ and $(a, b, \delta) = (1, 1, 0.5)$.

Intuitively, a greater prevalence of individuals with low attention budgets leads the

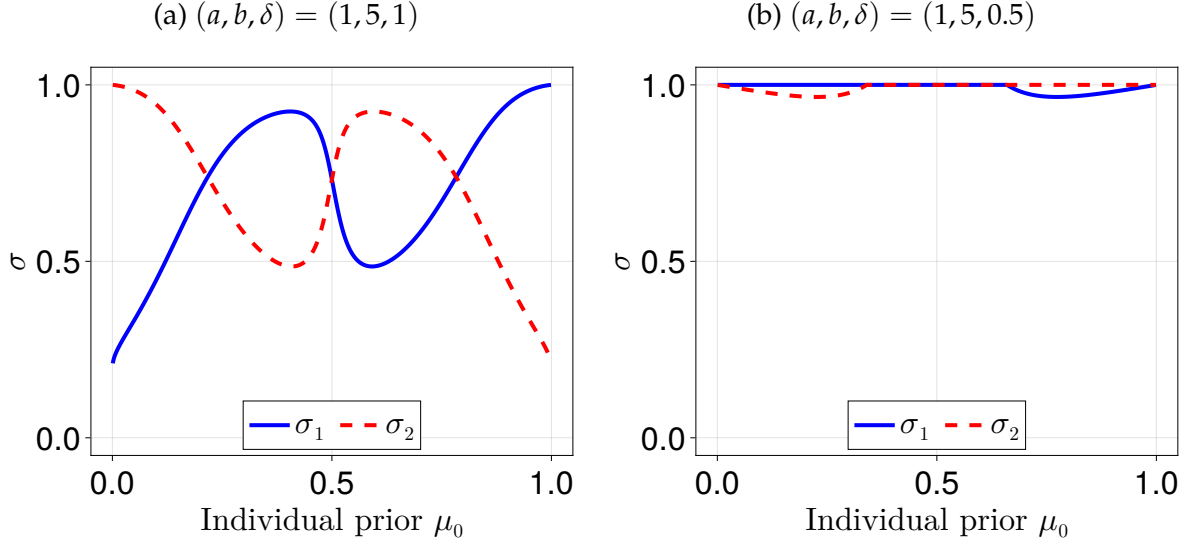


Figure 6: Optimal Message Precision Across Individual Priors by $(1, 5, \delta)$

central bank to optimally choose less precise messages in order to maintain broader engagement. This intuition is indeed borne out in the comparison of Figures 2a and 6a for an economy in which all individuals are inattentive ($\delta = 1$): as the distributional parameters shift from $(a, b) = (1, 1)$ to $(a, b) = (1, 5)$ —indicating greater mass concentrated on low attention budgets—the central bank sends less precise messages, reflected in overall lower values of σ_1 and σ_2 .

However, when only half of the individuals are inattentive (i.e., $\delta = 0.5$), the pattern reverses. As the attention budget distribution becomes increasingly right-skewed—with more mass concentrated on low attention budgets—the central bank finds it optimal instead to send more precise messages, as reflected in higher values of σ_1 and σ_2 —see the comparison between Figure 2b and Figure 6b. This is because the central bank considers not only the extent to which inattentive individuals engage with its communication, but also whom it chooses to target—that is, whether to prioritize attentive or inattentive individuals.

When the distribution of attention budgets among inattentive individuals is heavily right-skewed toward low budgets, and a substantial share of fully attentive individuals exists, the central bank may instead choose to communicate at the highest level of message precision, targeting only attentive individuals. In doing so, it deliberately forgoes engagement with inattentive individuals altogether. By strategically selecting the target audience for its communication, the central bank can achieve more effective macroeconomic stabilization at the aggregate level.

6 Conclusion

This paper studies the optimal information design of a central bank with commitment. We show that strategically crafted communication can serve as a powerful policy instrument for macroeconomic stabilization. Specifically, the central bank adjusts its information provision depending on whether the inflation surprise is excessive relative to the underlying unemployment shock. When inflation surprises are needed to counteract large shocks, the central bank optimally remains deliberately vague. Otherwise, it communicates informatively, balancing precision with audience reach to better manage individuals' inflation expectations.

Our results offer a novel interpretation of the central bank's deliberately vague communication. During a period of severe economic distress, the central bank may face a substantial unemployment shock. In such a context, preserving inflation surprises through vagueness allowed the central bank to stabilize the economy more effectively. In contrast, fully revealing the state of the economy would have neutralized the inflation surprise, reducing its capacity to counteract the shock.

We leave the coordinated design of monetary and communication policies for future research. In principle, a central bank could jointly optimize both instruments to improve stabilization in response to economic fluctuations. In such a framework, the relative weight assigned to inflation and unemployment gaps in the central bank's objective could affect the optimal mix of monetary policy and communication. When the effectiveness of inflation surprises declines, unconventional monetary tools may enhance policy flexibility and strengthen the role of communication. In addition, uncertainty about policy effects may induce Brainard-type attenuation, leading to more cautious policy responses and altering both the optimal precision of communication and the joint design of monetary and communication policies.

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A Mathematical Details

A.1 Proof of Proposition 1

We denote with $\sigma_1 = \sigma(s_1|\omega_1)$ and $\sigma_2 = \sigma(s_2|\omega_2)$. It follows that:

$$\mu_1 = \frac{\sigma_1 \mu_0}{\sigma_1 \mu_0 + (1 - \sigma_2)(1 - \mu_0)} \quad (23)$$

$$\mu_2 = \frac{(1 - \sigma_1) \mu_0}{(1 - \sigma_1) \mu_0 + \sigma_2(1 - \mu_0)} \quad (24)$$

$$\frac{\partial \mu_1}{\partial \sigma_1} = \frac{\mu_0(1 - \mu_0)(1 - \sigma_2)}{[\sigma_1 \mu_0 + (1 - \sigma_2)(1 - \mu_0)]^2} \quad (25)$$

$$\frac{\partial \mu_2}{\partial \sigma_1} = \frac{-\mu_0(1 - \mu_0)\sigma_2}{[(1 - \sigma_1) \mu_0 + \sigma_2(1 - \mu_0)]^2} \quad (26)$$

$$\frac{\partial \mu_1}{\partial \sigma_2} = \frac{\mu_0(1 - \mu_0)\sigma_1}{[\sigma_1 \mu_0 + (1 - \sigma_2)(1 - \mu_0)]^2} \quad (27)$$

$$\frac{\partial \mu_2}{\partial \sigma_2} = \frac{-\mu_0(1 - \mu_0)(1 - \sigma_1)}{[(1 - \sigma_1) \mu_0 + \sigma_2(1 - \mu_0)]^2} \quad (28)$$

$$\tau_1 = \sigma_1 \mu_0 + (1 - \sigma_2)(1 - \mu_0) \quad (29)$$

$$\tau_2 = (1 - \sigma_1) \mu_0 + \sigma_2(1 - \mu_0) \quad (30)$$

$$\frac{\partial \tau_j}{\partial \sigma_1} = \begin{cases} \mu_0 & \text{if } j = 1 \\ -\mu_0 & \text{otherwise} \end{cases} \quad (31)$$

$$\frac{\partial \tau_j}{\partial \sigma_2} = \begin{cases} 1 - \mu_0 & \text{if } j = 2 \\ -(1 - \mu_0) & \text{otherwise} \end{cases} \quad (32)$$

$$\frac{\partial H(\mu_j)}{\partial \sigma_k} = -\frac{\partial \mu_j}{\partial \sigma_k} \ln \left(\frac{\mu_j}{1 - \mu_j} \right) \quad (33)$$

$$c'_k(\sigma) = -\chi \sum_{j=1,2} \left[\frac{\partial \tau_j}{\partial \sigma_k} H(\mu_j) + \tau_j \frac{\partial H(\mu_j)}{\partial \sigma_k} \right] \quad (34)$$

Under Assumption 3, the problem in (10) becomes:

$$\begin{aligned} \min_{\sigma} \delta F(c(\sigma)) & \left\{ \mu_0^c [\omega_1 - 2\gamma\nu(1 - \mu_0)]^2 + (1 - \mu_0^c) [\omega_2 + 2\gamma\nu\mu_0]^2 \right\} + \\ & + [1 - \delta F(c(\sigma))] \left\{ \mu_0^c \left[\sigma_1(\omega_1 - 2\gamma\nu(1 - \mu_1))^2 + (1 - \sigma_1)(\omega_1 - 2\gamma\nu(1 - \mu_2))^2 \right] + \right. \\ & \left. + (1 - \mu_0^c) \left[\sigma_2(\omega_2 + 2\gamma\nu\mu_2)^2 + (1 - \sigma_2)(\omega_2 + 2\gamma\nu\mu_1)^2 \right] \right\} \end{aligned}$$

The F.O.C. are:

$$\begin{aligned}
& \delta f(c(\sigma))c'_1(\sigma) \left\{ \mu_0^\epsilon [\omega_1 - 2\gamma\nu(1 - \mu_0)]^2 + (1 - \mu_0^\epsilon) [\omega_2 + 2\gamma\nu\mu_0]^2 \right\} + \\
& -\delta f(c(\sigma))c'_1(\sigma) \left\{ \mu_0^\epsilon \left[\sigma_1(\omega_1 - 2\gamma\nu(1 - \mu_1))^2 + (1 - \sigma_1)(\omega_1 - 2\gamma\nu(1 - \mu_2))^2 \right] + \right. \\
& \quad \left. + (1 - \mu_0^\epsilon) \left[\sigma_2(\omega_2 + 2\gamma\nu\mu_2)^2 + (1 - \sigma_2)(\omega_2 + 2\gamma\nu\mu_1)^2 \right] \right\} + \\
& + [1 - \delta F(c(\sigma))] \left\{ \mu_0^\epsilon \left[(\omega_1 - 2\gamma\nu(1 - \mu_1))^2 + 4\sigma_1(\omega_1 - 2\gamma\nu(1 - \mu_1))\gamma\nu \frac{\partial \mu_1}{\partial \sigma_1} + \right. \right. \\
& \quad \left. - (\omega_1 - 2\gamma\nu(1 - \mu_2))^2 + 4(1 - \sigma_1)(\omega_1 - 2\gamma\nu(1 - \mu_2))\gamma\nu \frac{\partial \mu_2}{\partial \sigma_1} \right] + \\
& \quad \left. + 4\gamma\nu(1 - \mu_0^\epsilon) \left[\sigma_2(\omega_2 + 2\gamma\nu\mu_2) \frac{\partial \mu_2}{\partial \sigma_1} + (1 - \sigma_2)(\omega_2 + 2\gamma\nu\mu_1) \frac{\partial \mu_1}{\partial \sigma_1} \right] \right\} = 0 \\
& \delta f(c(\sigma))c'_2(\sigma) \left\{ \mu_0^\epsilon [\omega_1 - 2\gamma\nu(1 - \mu_0)]^2 + (1 - \mu_0^\epsilon) [\omega_2 + 2\gamma\nu\mu_0]^2 \right\} + \\
& -\delta f(c(\sigma))c'_2(\sigma) \left\{ \mu_0^\epsilon \left[\sigma_1(\omega_1 - 2\gamma\nu(1 - \mu_1))^2 + (1 - \sigma_1)(\omega_1 - 2\gamma\nu(1 - \mu_2))^2 \right] + \right. \\
& \quad \left. + (1 - \mu_0^\epsilon) \left[\sigma_2(\omega_2 + 2\gamma\nu\mu_2)^2 + (1 - \sigma_2)(\omega_2 + 2\gamma\nu\mu_1)^2 \right] \right\} + \\
& + [1 - \delta F(c(\sigma))] \left\{ (1 - \mu_0^\epsilon) \left[(\omega_2 + 2\gamma\nu\mu_2)^2 + 4\sigma_2(\omega_2 + 2\gamma\nu\mu_2)\gamma\nu \frac{\partial \mu_2}{\partial \sigma_2} + \right. \right. \\
& \quad \left. - (\omega_2 + 2\gamma\nu\mu_1)^2 + 4(1 - \sigma_2)(\omega_2 + 2\gamma\nu\mu_1)\gamma\nu \frac{\partial \mu_1}{\partial \sigma_2} \right] + \\
& \quad \left. + 4\gamma\nu\mu_0^\epsilon \left[\sigma_1(\omega_1 - 2\gamma\nu(1 - \mu_1)) \frac{\partial \mu_1}{\partial \sigma_2} + (1 - \sigma_1)(\omega_1 - 2\gamma\nu(1 - \mu_2)) \frac{\partial \mu_2}{\partial \sigma_2} \right] \right\} = 0
\end{aligned}$$

Under Assumptions 1-2, the F.O.C.s becomes symmetric. Thus, we consider symmetric solutions where $\sigma_1 = \sigma_2 = \sigma^*$ and $\sigma^* \in \left[\frac{1}{2}, 1\right]$ is the precision of the central bank's information. It follows that the solution to the central bank's information design problem solves the following condition:

$$2\gamma\nu(4\omega - 2\gamma\nu) \left\{ [1 - \delta F(c(\sigma))](2\sigma^* - 1) + 2\delta f(c(\sigma))c'(\sigma) \left[\sigma^*(1 - \sigma^*) - \frac{1}{4} \right] \right\} = 0$$

Removing irrelevant positive terms yields:

$$(2\omega - \gamma\nu) \left\{ [1 - \delta F(c(\sigma))] \left(\sigma^* - \frac{1}{2} \right) + \delta f(c(\sigma)) c'(\sigma) \left[\sigma^*(1 - \sigma^*) - \frac{1}{4} \right] \right\} = 0$$

Observe that:

$$\sigma^*(1 - \sigma^*) - \frac{1}{4} = - \left(\sigma^* - \frac{1}{2} \right)^2$$

and that $1 - \delta F(c(\sigma)) > 0$ for any σ and $\delta \in [0, 1]$. Substituting the quadratic expression and dividing by $1 - \delta F(c(\sigma))$ yields:

$$(2\omega - \gamma\nu) \left\{ \left(\sigma^* - \frac{1}{2} \right) - \frac{\delta f(c(\sigma))}{1 - \delta F(c(\sigma))} c'(\sigma) \left(\sigma^* - \frac{1}{2} \right)^2 \right\} = 0$$

Observe that:

$$\frac{d}{d\sigma} \ln(1 - \delta F(c(\sigma))) = - \frac{\delta f(c(\sigma))}{1 - \delta F(c(\sigma))} c'(\sigma) \leq 0$$

This implies that:

$$(2\omega - \gamma\nu) \left(\sigma^* - \frac{1}{2} \right) \left[1 + \left(\frac{d}{d\sigma} \ln(1 - \delta F(c(\sigma))) \right) \left(\sigma^* - \frac{1}{2} \right) \right] = 0$$

Under Assumption 4, F denotes the CDF for a uniform distribution on $[0, 1]$:

$$F(c(\sigma)) = c(\sigma)$$

This implies:

$$\ln(1 - \delta F(c(\sigma))) = \ln(1 - \delta c(\sigma))$$

Accordingly, two possible candidates satisfy the necessary condition: $\underline{\sigma} = \frac{1}{2}$ and $\bar{\sigma} \in (\frac{1}{2}, 1]$ such that:

$$1 + \left(\frac{d}{d\sigma} \ln(1 - \delta c(\sigma)) \right) \Big|_{\sigma=\bar{\sigma}} \left(\bar{\sigma} - \frac{1}{2} \right) = 0$$

or $\bar{\sigma} = 1$ if

$$1 + \frac{1}{2} \left(\frac{d}{d\sigma} \ln(1 - \delta c(\sigma)) \right) \Big|_{\sigma=1} > 0$$

Therefore $\sigma^* \in \{\underline{\sigma}, \bar{\sigma}\}$. To show which candidate is sufficient, we compare utilities across them. Central bank's utility with a generic $\hat{\sigma}$ is:

$$\begin{aligned} u_{\hat{\sigma}} = & -\delta F(c(\sigma))(\omega - \gamma\nu)^2 \\ & - (1 - \delta F(c(\sigma))) \left[\hat{\sigma}(\omega - 2\gamma\nu(1 - \hat{\sigma}))^2 + (1 - \hat{\sigma})(\omega - 2\gamma\nu\hat{\sigma})^2 \right] \end{aligned}$$

Its utility with uninformative communication $\underline{\sigma}$ is:

$$u_{\underline{\sigma}} = -(\omega - \gamma\nu)^2$$

For $\sigma^* = \underline{\sigma}$ to be optimal, $u_{\underline{\sigma}} - u_{\hat{\sigma}} \geq 0$. Under which condition does this relationship hold?

In the first step, $u_{\hat{\sigma}}$ can be rewritten into:

$$\begin{aligned} u_{\hat{\sigma}} &= -\delta F(c(\sigma))(\omega - \gamma\nu)^2 \\ &\quad - (1 - \delta F(c(\sigma)))(\omega - \gamma\nu)^2 + (1 - \delta F(c(\sigma)))(\omega - \gamma\nu)^2 \\ &\quad - (1 - \delta F(c(\sigma))) \left[\hat{\sigma}(\omega - 2\gamma\nu(1 - \hat{\sigma}))^2 + (1 - \hat{\sigma})(\omega - 2\gamma\nu\hat{\sigma})^2 \right] \end{aligned}$$

It follows that:

$$u_{\hat{\sigma}} = u_{\underline{\sigma}} - (1 - \delta F(c(\sigma))) \left[\hat{\sigma}(\omega - 2\gamma\nu(1 - \hat{\sigma}))^2 + (1 - \hat{\sigma})(\omega - 2\gamma\nu\hat{\sigma})^2 - (\omega - \gamma\nu)^2 \right]$$

Thus, showing $u_{\underline{\sigma}} - u_{\hat{\sigma}} \geq 0$ is equivalent to show:

$$\hat{\sigma}(\omega - 2\gamma\nu(1 - \hat{\sigma}))^2 + (1 - \hat{\sigma})(\omega - 2\gamma\nu\hat{\sigma})^2 - (\omega - \gamma\nu)^2 \geq 0$$

In the second step,

$$\begin{aligned} \hat{\sigma}(\omega - 2\gamma\nu(1 - \hat{\sigma}))^2 &= \hat{\sigma}\omega^2 - 4\omega\gamma\nu(1 - \hat{\sigma})\hat{\sigma} + 4\gamma^2\nu^2(1 - \hat{\sigma})^2\hat{\sigma} \\ (1 - \hat{\sigma})(\omega - 2\gamma\nu\hat{\sigma})^2 &= (1 - \hat{\sigma})\omega^2 - 4\omega\gamma\nu(1 - \hat{\sigma})\hat{\sigma} + 4\gamma^2\nu^2(1 - \hat{\sigma})\hat{\sigma}^2 \\ (\omega - \gamma\nu)^2 &= \omega^2 - 2\omega\gamma\nu + \gamma^2\nu^2 \end{aligned}$$

The above inequality can be rearranged as:

$$-8\omega\gamma\nu(1 - \hat{\sigma})\hat{\sigma} + 4\gamma^2\nu^2(1 - \hat{\sigma})\hat{\sigma} + 2\omega\gamma\nu - \gamma^2\nu^2 \geq 0$$

It can be further expressed nicely as:

$$\begin{aligned} -4\gamma\nu(1 - \hat{\sigma})\hat{\sigma}(2\omega - \gamma\nu) + \gamma\nu(2\omega - \gamma\nu) &\geq 0 \\ \gamma\nu(2\omega - \gamma\nu) [1 - 4(1 - \hat{\sigma})\hat{\sigma}] &\geq 0 \end{aligned}$$

Given that $\gamma\nu > 0$ by construction and $1 - 4(1 - \hat{\sigma})\hat{\sigma} \geq 0$ for any $\hat{\sigma} \in [\frac{1}{2}, 1]$, we find that, consistent with the necessary condition, $\sigma^* = \underline{\sigma} = \frac{1}{2}$ if $2\omega - \gamma\nu \geq 0$ while $\sigma^* = \bar{\sigma} \in (\frac{1}{2}, 1]$ if $2\omega - \gamma\nu < 0$.

A.2 Proof of Lemma 2

Proof.

$$\begin{aligned}
\pi^e(\hat{\mu}) &= \pi^T - \nu(1 - 2\hat{\mu}) \\
&= \pi^T - \nu \{1 - 2[(1 - \theta)\mu_0 + \theta\mu]\} \\
&= \pi^T + 2\nu(1 - \theta)\mu_0 + 2\nu\theta\mu - \nu \\
&= (1 - \theta)\pi^T + \theta\pi^T + 2\nu(1 - \theta)\mu_0 + 2\nu\theta\mu - (1 - \theta)\nu - \theta\nu \\
&= (1 - \theta) [\pi^T - \nu(1 - 2\mu_0)] + \theta [\pi^T - \nu(1 - 2\mu)] \\
&= (1 - \theta)\pi^e(\mu_0) + \theta\pi^e(\mu)
\end{aligned}$$

□

A.3 Proof of Proposition 2

Under adaptive belief, the problem in (10) becomes:

$$\begin{aligned}
\min_{\sigma} \delta F(c(\sigma)) &\left\{ \mu_0^c [\omega_1 - 2\gamma\nu(1 - \mu_0)]^2 + (1 - \mu_0^c) [\omega_2 + 2\gamma\nu\mu_0]^2 \right\} + \\
&+ [1 - \delta F(c(\sigma))] \left\{ \mu_0^c \left[\sigma_1 (\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_1)))^2 + \right. \right. \\
&+ (1 - \sigma_1) (\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_2)))^2 \Big] + \\
&+ (1 - \mu_0^c) \left[\sigma_2 (\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_2)))^2 + \right. \\
&+ (1 - \sigma_2) (\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_1)))^2 \Big] \Big\}
\end{aligned}$$

The F.O.C. are:

$$\begin{aligned}
& \delta f(c(\sigma))c'_1(\sigma) \left\{ \mu_0^\epsilon [\omega_1 - 2\gamma\nu(1 - \mu_0)]^2 + (1 - \mu_0^\epsilon) [\omega_2 + 2\gamma\nu\mu_0]^2 \right\} + \\
& -\delta f(c(\sigma))c'_1(\sigma) \left\{ \mu_0^\epsilon \left[\sigma_1(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_1)))^2 + \right. \right. \\
& \quad \left. \left. + (1 - \sigma_1)(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_2)))^2 \right] + \right. \\
& \quad \left. + (1 - \mu_0^\epsilon) \left[\sigma_2(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_2)))^2 + \right. \right. \\
& \quad \left. \left. + (1 - \sigma_2)(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_1)))^2 \right] \right\} + \\
& + [1 - \delta F(c(\sigma))] \left\{ \mu_0^\epsilon \left[(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_1)))^2 + \right. \right. \\
& \quad \left. \left. + 4\sigma_1(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_1)))\gamma\theta\nu\frac{\partial\mu_1}{\partial\sigma_1} + \right. \right. \\
& \quad \left. \left. - (\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_2)))^2 + \right. \right. \\
& \quad \left. \left. + 4(1 - \sigma_1)(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_2)))\gamma\theta\nu\frac{\partial\mu_2}{\partial\sigma_1} \right] + \right. \\
& \quad \left. + 4\gamma\theta\nu(1 - \mu_0^\epsilon) \left[\sigma_2(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_2)))\frac{\partial\mu_2}{\partial\sigma_1} + \right. \right. \\
& \quad \left. \left. + (1 - \sigma_2)(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_1)))\frac{\partial\mu_1}{\partial\sigma_1} \right] \right\} = 0
\end{aligned}$$

$$\begin{aligned}
& \delta f(c(\sigma))c'_2(\sigma) \left\{ \mu_0^\epsilon [\omega_1 - 2\gamma\nu(1 - \mu_0)]^2 + (1 - \mu_0^\epsilon) [\omega_2 + 2\gamma\nu\mu_0]^2 \right\} + \\
& -\delta f(c(\sigma))c'_2(\sigma) \left\{ \mu_0^\epsilon \left[\sigma_1(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_1)))^2 + \right. \right. \\
& \quad \left. \left. + (1 - \sigma_1)(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_2)))^2 \right] + \right. \\
& \quad \left. + (1 - \mu_0^\epsilon) \left[\sigma_2(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_2)))^2 + \right. \right. \\
& \quad \left. \left. + (1 - \sigma_2)(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_1)))^2 \right] \right\} + \\
& + [1 - \delta F(c(\sigma))] \left\{ (1 - \mu_0^\epsilon) \left[(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_2)))^2 + \right. \right. \\
& \quad 4\sigma_2(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_2)))\gamma\theta\nu\frac{\partial\mu_2}{\partial\sigma_2} + \\
& \quad \left. - (\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_1)))^2 + \right. \\
& \quad \left. 4(1 - \sigma_2)(\omega_2 + \gamma\nu(1 - (1 - \theta)(1 - 2\mu_0) - \theta(1 - 2\mu_1)))\gamma\theta\nu\frac{\partial\mu_1}{\partial\sigma_2} \right] + \\
& \quad + 4\gamma\theta\nu\mu_0^\epsilon \left[\sigma_1(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_1)))\frac{\partial\mu_1}{\partial\sigma_2} + \right. \\
& \quad \left. (1 - \sigma_1)(\omega_1 - \gamma\nu(1 + (1 - \theta)(1 - 2\mu_0) + \theta(1 - 2\mu_2)))\frac{\partial\mu_2}{\partial\sigma_2} \right] \left. \right\} = 0
\end{aligned}$$

Applying assumptions **1-2**, the F.O.C.s becomes symmetric. Thus, we consider symmetric solutions where $\sigma_1 = \sigma_2 = \sigma^*$. It follows that the solution to the central bank's information design problem solves the following condition:

$$2\theta\gamma\nu(4\omega - 2(2 - \theta)\gamma\nu) \left\{ [1 - \delta F(c(\sigma))](2\sigma^* - 1) + 2\delta f(c(\sigma))c'(\sigma) \left[\sigma^*(1 - \sigma^*) - \frac{1}{4} \right] \right\} = 0$$

Removing irrelevant positive terms yields:

$$(2\omega - (2 - \theta)\gamma\nu) \left\{ [1 - \delta F(c(\sigma))] \left(\sigma^* - \frac{1}{2} \right) + \delta f(c(\sigma))c'(\sigma) \left[\sigma^*(1 - \sigma^*) - \frac{1}{4} \right] \right\} = 0$$

Observe that:

$$\sigma^*(1 - \sigma^*) - \frac{1}{4} = - \left(\sigma^* - \frac{1}{2} \right)^2$$

and that $1 - \delta F(c(\sigma)) > 0$ for any σ and $\delta \in [0, 1]$. Substituting the quadratic expres-

sion and dividing by $1 - F(c(\sigma))$ yields:

$$(2\omega - (2 - \theta)\gamma\nu) \left\{ \left(\sigma^* - \frac{1}{2} \right) - \frac{\delta f(c(\sigma))}{1 - \delta F(c(\sigma))} c'(\sigma) \left(\sigma^* - \frac{1}{2} \right)^2 \right\} = 0$$

Observe that:

$$\frac{d}{d\sigma} \ln(1 - \delta F(c(\sigma))) = -\frac{\delta f(c(\sigma))}{1 - \delta F(c(\sigma))} c'(\sigma) \leq 0$$

This implies that:

$$(2\omega - (2 - \theta)\gamma\nu) \left(\sigma^* - \frac{1}{2} \right) \left[1 + \left(\frac{d}{d\sigma} \ln(1 - \delta F(c(\sigma))) \right) \left(\sigma^* - \frac{1}{2} \right) \right] = 0$$

Accordingly, two possible candidates satisfy the necessary condition: $\underline{\sigma} = \frac{1}{2}$ and $\bar{\sigma} \in (\frac{1}{2}, 1]$ such that:

$$1 + \left(\frac{d}{d\sigma} \ln(1 - \delta F(c(\sigma))) \right) \Big|_{\sigma=\bar{\sigma}} \left(\bar{\sigma} - \frac{1}{2} \right) = 0.$$

or $\bar{\sigma} = 1$ if

$$1 + \frac{1}{2} \left(\frac{d}{d\sigma} \ln(1 - \delta F(c(\sigma))) \right) \Big|_{\sigma=1} > 0$$

Therefore, $\sigma^* \in \{\underline{\sigma}, \bar{\sigma}\}$. To show which candidate is sufficient, we compare utilities across them. Central bank's utility with a generic $\hat{\sigma}$ is:

$$u_{\hat{\sigma}} = -\delta F(c(\sigma))(\omega - \gamma\nu)^2 - (1 - \delta F(c(\sigma))) \left[\hat{\sigma}(\omega - \gamma\nu(1 + \theta(1 - 2\hat{\sigma})))^2 + (1 - \hat{\sigma})(\omega - \gamma\nu(1 - \theta(1 - 2\hat{\sigma})))^2 \right]$$

Its utility with uninformative communication $\underline{\sigma}$ is:

$$u_{\underline{\sigma}} = -(\omega - \gamma\nu)^2$$

For $\sigma^* = \underline{\sigma}$ to be optimal, $u_{\underline{\sigma}} - u_{\hat{\sigma}} \geq 0$. Under which condition does this relationship hold?

In the first step, $u_{\hat{\sigma}}$ can be rewritten into:

$$u_{\hat{\sigma}} = -\delta F(c(\sigma))(\omega - \gamma\nu)^2 - (1 - \delta F(c(\sigma)))(\omega - \gamma\nu)^2 + (1 - \delta F(c(\sigma)))(\omega - \gamma\nu)^2 - (1 - \delta F(c(\sigma))) \left[\hat{\sigma}(\omega - \gamma\nu(1 + \theta(1 - 2\hat{\sigma})))^2 + (1 - \hat{\sigma})(\omega - \gamma\nu(1 - \theta(1 - 2\hat{\sigma})))^2 \right]$$

It follows that:

$$u_{\hat{\sigma}} = u_{\underline{\sigma}} - (1 - \delta F(c(\sigma))) \left[\hat{\sigma}(\omega - \gamma\nu(1 + \theta(1 - 2\hat{\sigma})))^2 + (1 - \hat{\sigma})(\omega - \gamma\nu(1 - \theta(1 - 2\hat{\sigma})))^2 - (\omega - \gamma\nu)^2 \right]$$

Thus, showing $u_{\underline{\sigma}} - u_{\hat{\sigma}} \geq 0$ is equivalent to show:

$$\hat{\sigma}(\omega - \gamma\nu(1 + \theta(1 - 2\hat{\sigma})))^2 + (1 - \hat{\sigma})(\omega - \gamma\nu(1 - \theta(1 - 2\hat{\sigma})))^2 - (\omega - \gamma\nu)^2 \geq 0$$

In the second step,

$$\begin{aligned} \hat{\sigma}(\omega - \gamma\nu(1 + \theta(1 - 2\hat{\sigma})))^2 &= \hat{\sigma}[\omega^2 - 2\omega\gamma\nu(1 + \theta(1 - 2\hat{\sigma})) + \gamma^2\nu^2(1 + \theta(1 - 2\hat{\sigma}))^2] \\ (1 - \hat{\sigma})(\omega - \gamma\nu(1 - \theta(1 - 2\hat{\sigma})))^2 &= (1 - \hat{\sigma})[\omega^2 - 2\omega\gamma\nu(1 - \theta(1 - 2\hat{\sigma})) + \gamma^2\nu^2(1 - \theta(1 - 2\hat{\sigma}))^2] \\ (\omega - \gamma\nu)^2 &= \omega^2 - 2\omega\gamma\nu + \gamma^2\nu^2 \end{aligned}$$

The above inequality can be rearranged as:

$$-2\omega\gamma\nu[1 - \theta(1 - 2\hat{\sigma})^2] + \gamma^2\nu^2[1 + \theta^2(1 - 2\hat{\sigma})^2 - 2\theta(1 - 2\hat{\sigma})^2] + 2\omega\gamma\nu - \gamma^2\nu^2 \geq 0$$

It can be further expressed nicely as:

$$\begin{aligned} 2\omega\gamma\nu\theta(1 - 2\hat{\sigma})^2 + \gamma^2\nu^2\theta(\theta - 2)(1 - 2\hat{\sigma})^2 &\geq 0 \\ \gamma\nu\theta(1 - 2\hat{\sigma})^2[2\omega - \gamma\nu(2 - \theta)] &\geq 0 \end{aligned}$$

Given that $\gamma\nu\theta \geq 0$ by construction and $(1 - 2\hat{\sigma})^2 \geq 0$ for any $\hat{\sigma} \in [\frac{1}{2}, 1]$, we find that, consistent with the necessary condition, $\sigma^* = \underline{\sigma} = \frac{1}{2}$ if $2\omega - \gamma\nu(2 - \theta) \geq 0$ while $\sigma^* = \bar{\sigma} \in (\frac{1}{2}, 1]$ if $2\omega - \gamma\nu(2 - \theta) < 0$.

B Extension to Shock Magnitude and Asymmetry

In reality, shocks are not always moderate or symmetric. Episodes such as financial crises, pandemics, or geopolitical disruptions often involve disturbances that are either unprecedented in scale or asymmetric in nature. During such periods, individuals and the central bank are likely to hold misaligned economic outlooks or divergent prior beliefs about underlying states. These conditions challenge the conclusions derived under Assumptions 1 and 2, which presume neutral beliefs and moderately symmetric disturbances, and underscore the need for a systematic reevaluation of how the central bank should communicate under belief heterogeneity and economic turbulence.

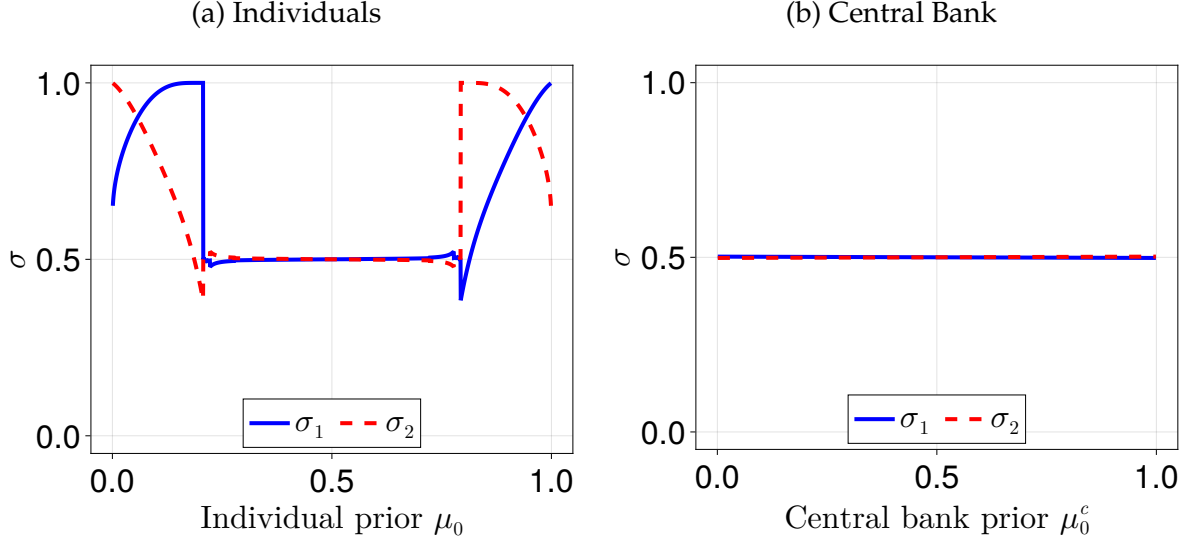


Figure 7: Optimal Message Precision Across Priors when $(\omega_1, \omega_2) = (3, -3)$

To this end, we examine the role of belief heterogeneity and asymmetric unemployment shocks in shaping central bank communication by jointly relaxing Assumptions 1 and 2. Specifically, we explore the optimal communication strategy across varying prior beliefs of both individuals and the central bank, ranging from 0 to 1. Meanwhile, we consider two alternative shock configurations in addition to the benchmark: a large symmetric shock scenario $(\omega_1, \omega_2) = (3, -3)$ and an asymmetric shock scenario $(\omega_1, \omega_2) = (3, -1)$. All other parameters are held at their benchmark values. Recall that the benchmark case corresponds to $(\omega_1, \omega_2) = (1, -1)$. In both counterfactual scenarios, one or both shocks are three times larger in magnitude than in the benchmark, allowing us to assess how the central bank adjusts its communication strategy under more volatile or imbalanced economic conditions.

We present the results under the large symmetric shock scenario $(\omega_1, \omega_2) = (3, -3)$ in Figure 7. Figure 7a displays the optimal communication strategy as individual prior beliefs vary, while Figure 7b illustrates the case in which the central bank's prior belief varies with individual priors held fixed.

Under neutral individual beliefs, the large symmetric shock scenario leads the central bank to optimally choose vague communication, as implied by Proposition 1.¹⁷ When considering heterogeneous individual prior beliefs alongside increased shock magnitude, Figure 7a reveals the emergence of a strategic vagueness region—a middle range of priors over which the central bank finds it optimal to remain uninformative, thereby leveraging full-scale inflation surprises to counteract large disturbances. According to the fundamental comparison in Proposition 1, when individual prior be-

¹⁷The condition reads: $2\omega \geq \gamma\nu$. Under the large symmetric shock scenario, this inequality holds as $2 \cdot 3 \geq 2.94 \cdot 1$.

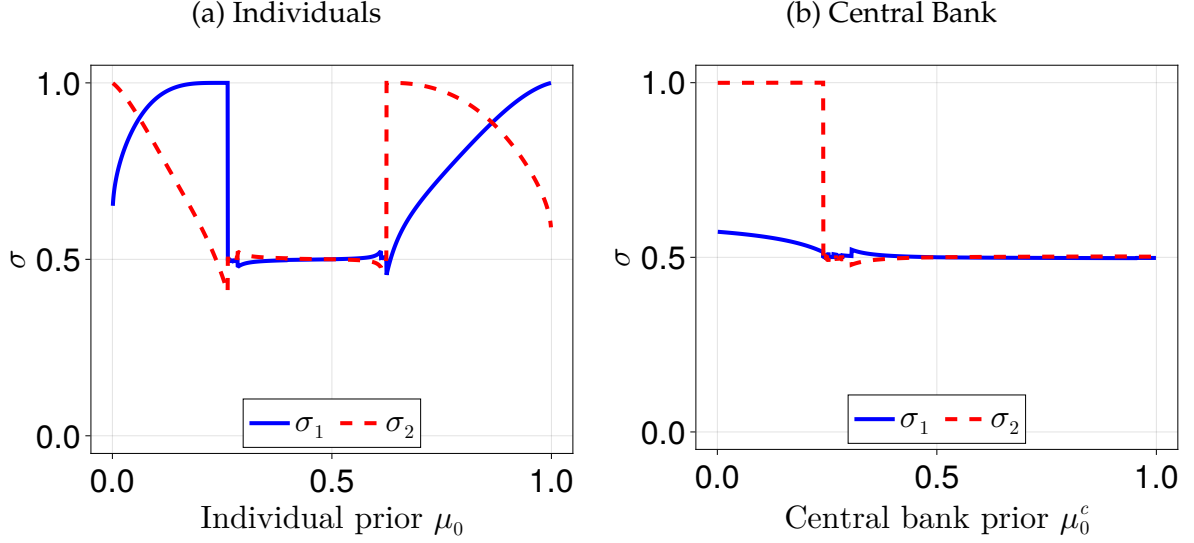


Figure 8: Optimal Message Precision Across Priors when $(\omega_1, \omega_2) = (3, -1)$

liefs are relatively neutral, the shock magnitude dominates the effective power of inflation surprises, prompting the central bank to refrain from informative communication. However, as individuals become sufficiently pessimistic or optimistic, this comparison reverses: the effective power of inflation surprises outweighs the shock magnitude, encouraging the central bank to communicate informatively.

This result closely parallels the case of a flat Phillips curve with $\gamma = 1$, as shown in Figure 4a, where a reduced γ weakens the effective power of inflation surprises and generates a comparable strategic vagueness region. See the discussion accompanying Figure 4a for further reference.

Conversely, Figure 7b shows that central bank prior belief plays no role in shaping communication under the large symmetric shock scenario. As discussed in Figure 5a, varying the central bank's prior simply reweights two equivalent state-contingent losses in the objective function and therefore does not affect the optimal communication strategy. See the discussion accompanying Figure 5a for further reference.

We next present the results under the asymmetric shock scenario $(\omega_1, \omega_2) = (3, -1)$ in Figure 8. Figure 8a displays the optimal communication strategy as individual prior beliefs vary, while Figure 8b illustrates the case in which the central bank's prior belief varies, holding individual priors fixed.

As in the large symmetric shock scenario shown in Figure 7a, the strategic vagueness region also arises under the asymmetric shock scenario, as illustrated in Figure 8a. However, the shape and extent of this region differ between the two settings. Under symmetric shocks, the region is symmetric and centered around the neutral individual prior. In contrast, with asymmetric shocks, the region becomes skewed due to the

imbalance in shock magnitudes. Specifically, it shifts toward more optimistic individual priors, given the stronger shock associated with the weak state ($\omega_1 > \omega_2$). This observation naturally raises the question: why does a stronger shock in the weak state cause the strategic vagueness region to extend disproportionately over more optimistic individual priors?

Although the fundamental comparison—between shock magnitude and the effective power of inflation surprises—is originally derived under the assumption of symmetric shocks, it remains informative for analyzing asymmetric scenarios. As previously discussed, inflation surprises intensify in the state that individuals perceive as less likely. In the present asymmetric setting, the relatively small shock magnitude in the strong state implies that, under heightened individual pessimism, inflation surprises in the strong state can quickly exceed the magnitude of the underlying shock, prompting the central bank to adopt informative communication at a relatively low threshold of pessimism. Conversely, the larger shock associated with the weak state allows the central bank to tolerate greater inflation surprises in that state as individual optimism increases, thereby widening the range of optimistic priors over which uninformative communication remains optimal.

Taken together, these dynamics induce an asymmetric response in communication: the central bank begins to communicate informatively earlier under pessimism than under optimism, reflecting the dominance of the weak-state shock magnitude over that of the strong state.

Figure 8b displays a hybrid pattern that combines elements from Figure 7b (large symmetric shocks) and Figure 3a (benchmark small symmetric shocks). Under the large symmetric shock scenario, the central bank optimally chooses to remain vague, leveraging large inflation surprises to counteract substantial shock disturbances. In contrast, under the benchmark scenario with small symmetric shocks, the central bank communicates informatively, tailoring its messages to correct inflation surprises in the state it deems more likely.

In the asymmetric shock scenario, the weak state is subject to a large shock, whereas the strong state remains associated with a smaller shock, as in the benchmark. This asymmetry implies that inflation surprises become relatively excessive in the strong state. In this context, the central bank favors uninformative communication under pessimistic or moderately optimistic prior beliefs. However, as the central bank's prior becomes sufficiently optimistic—or places greater weight on the strong state—it shifts toward informative communication, characterized by a high σ_2 and a moderate σ_1 , in order to primarily correct inflation surprises in the strong state—although in this way renouncing to leverage inflation surprises to slightly compensate for the shock in the

weak state.